

Solar N appendix

Sun · N · downloadable appendix

Solar nitrogen: 8.189 ± 0.050

Asplund, Amarsi & Grevesse 2021: 7.83 ± 0.07 · this measurement +0.359 dex

PRELIMINARY. This value sits +0.36 dex above the Asplund 2021 reference and is not yet a settled result. The cause is identified and is not the NLTE treatment: the N I NLTE correction is measured and small (-0.0115 / -0.0145 / -0.0154 dex per line, because N I red is near-LTE at the Sun), and the surviving offset is a gf / data-channel floor on the Kitt Peak red multiplets (RYA-161) that is owed curation. It is published because withholding a measured number is not the same as correcting it — but expect it to move.

Kitt Peak · red-optical · Reference Grade · SET-AGSS21_ENGINE-A · ENGINE-A · n = 2. The tightest admitted product; independent products and engines are not averaged together.

Highlighted Products

red-optical

8.189

±0.027 stat · ±0.042 syst

SET-AGSS21_ENGINE-A · N I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied · solar_kpno_kurucz2005_corrected · Reference Grade

NIR

7.999

±0.037 stat · ±0.033 syst

MOL-CN_AX_IR · N I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied · solar_iag · Codex Grade

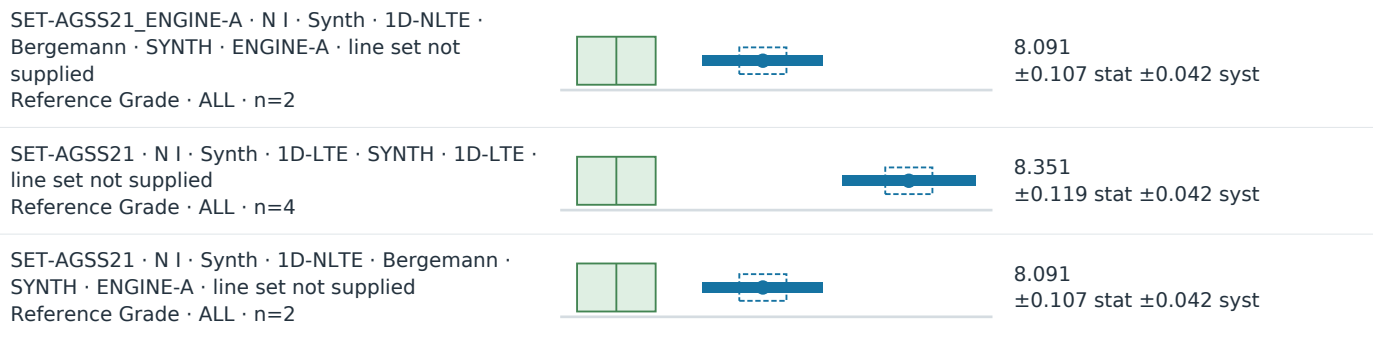
Error-bar Forest Plot

VIS — harps solar_harps_molecfits_corrected
A(N) dex; axis 6.52-8.25



VIS axis: 6.52 · 6.95 · 7.38 · 7.82 · 8.25 A(N) dex

red-optical — iag_fts_solar_atlas solar_iag
A(N) dex; axis 7.73-8.50



red-optical — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(N) dex; axis 7.73-8.50

SET-AGSS21_ENGINE-A · N I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Reference Grade · ALL · n=2



8.189
±0.027 stat ±0.042 syst

SET-AGSS21 · N I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=4



8.293
±0.069 stat ±0.042 syst

SET-AGSS21 · N I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=2



8.189
±0.027 stat ±0.042 syst

red-optical — kpno_solar_atlas solar_kpno_molecfit_corrected A(N) dex; axis 7.73-8.50

SET-AGSS21_ENGINE-A · N I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Reference Grade · ALL · n=2



8.016
±0.155 stat ±0.042 syst

SET-AGSS21 · N I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=4



8.287
±0.122 stat ±0.042 syst

SET-AGSS21 · N I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=2



8.016
±0.155 stat ±0.042 syst

red-optical axis: 7.73 · 7.92 · 8.12 · 8.31 · 8.50 A(N) dex

NIR — iag_fts_solar_atlas solar_iag A(N) dex; axis 7.73-8.31

MOL-CN_AX_IR · N I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=244



7.999
±0.037 stat ±0.033 syst

NIR — kpno_solar_atlas solar_kpno_molecfit_corrected A(N) dex; axis 7.73-8.31

MOL-CN_AX_IR · N I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=215



8.233
±0.046 stat ±0.036 syst

NIR axis: 7.73 · 7.87 · 8.02 · 8.16 · 8.31 A(N) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

How this was measured. One row per band. The indicator set is what was fitted; the holding is the solar spectrum it was fitted against; the engine and treatment say how. Dominant term is the largest single contributor to the reported uncertainty, and gf rung is the pedigree of the oscillator strengths that pool could reach.

Solar nitrogen context

The nitrogen strategy uses red-optical N I as the primary atomic route, with NH providing an independent molecular check and CN a carbon-dependent cross-check. An unresolved or unconstrained fit supplies no adopted Solar nitrogen abundance.

[C/O indicator strategy](#) · [Methodology](#) · [Nitrogen strategy](#)

Open provenance decisions

NIST's N I values ARE Tachiev (TP T7370). Adjudicating N I against this pull is one source seen twice, not independent confirmation.

[Adjudication record](#) · [Indicator gf checks](#)

Download the evidence

[Download PDF](#) — Solar N appendix

[Source feed](#) — includes quarantined fit outputs, not adopted abundances

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned CNO campaign snapshot. Source commit

22ea65678043e016b12fb0defa596c10a200df88

Feed N.json v1.14 · updated 2026-09-25T19:39:46Z

Generated 2026-09-25T21:53:06-06:00 (source commit timestamp)

Feed SHA-256 9e4244db27dd4f2683e3cca67fd49973b9da6ac535edf171744ad5a23ec32fe2

Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

[Science and provenance source notes](#)

Solar abundance and indicator analyses. Solar N atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [\[1\]](#), [\[2\]](#).

Atomic data and gf authorities. The recorded N I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [\[3\]](#), [\[4\]](#).

Atomic data and gf authorities. NIST's N I values ARE Tachiev (TP T7370). Adjudicating N I against this pull is one source seen twice, not independent confirmation. Sources: [\[3\]](#), [\[5\]](#).

Model and grid sources. The N I registry uses the Amarsi GALAH/PySME 1D-NLTE departure grid. The solar 3D-NLTE atomic analysis is a separate source, not a full-3D correction applied by this feed. Sources: [\[6\]](#), [\[7\]](#), [\[8\]](#).

Atomic data and gf authorities. Atomic N I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [\[3\]](#), [\[4\]](#), [\[5\]](#).

Model and grid sources. Feed label: Synth · 1D-NLTE · Bergemann. Registered source: Amarsi/PySME N I 1D-NLTE. The generic Bergemann label is retained as supplied and is not used as a bibliographic attribution. Sources: [6], [7], [8].

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [9].

Molecular line data and delivery. CN line-data delivery: CN red A-X: Brooke et al. 2014 / Sneden et al. 2014 (isotopologues). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [9], [10], [11].

Product-to-source index

N-products-92f4078d16e7 · products · red-optical · solar_iag · SET-AGSS21_ENGINE-A · ENGINE-A · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-b6f3d1679545 · products · red-optical · solar_iag · SET-AGSS21 · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

N-products-e3c293991e58 · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21_ENGINE-A · ENGINE-A · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-08261af2d8eb · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21 · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

N-products-e61a07d3c5f5 · products · red-optical · solar_kpno_molecfit_corrected · SET-AGSS21_ENGINE-A · ENGINE-A · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-dd0e12b67a86 · products · red-optical · solar_kpno_molecfit_corrected · SET-AGSS21 · 1D-LTE · not supplied · [1], [3], [4], [5], [9]

N-products-a673b561de48 · products · red-optical · solar_iag · SET-AGSS21 · ENGINE-A · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-e57946a904b1 · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21 · ENGINE-A · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-f95159b2157c · products · red-optical · solar_kpno_molecfit_corrected · SET-AGSS21 · ENGINE-A · not supplied · [1], [3], [4], [5], [6], [7], [8]

N-products-150d2f215a8b · products · VIS · solar_harps_molecfit_corrected · MOL-CN_red · 1D-LTE · not supplied · [1], [2], [9], [10], [11]

N-products-9944e2b77baf · products · NIR · solar_iag · MOL-CN_AX_IR · 1D-LTE · not supplied · [1], [2], [9], [10], [11]

N-products-7fee3da2024e · products · NIR · solar_kpno_molecfit_corrected · MOL-CN_AX_IR · 1D-LTE · not supplied · [1], [2], [9], [10], [11]

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

[2] — Amarsi et al. 2021, A&A 656 A113 DOI: 10.1051/0004-6361/202141384. [source record](#)

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. [source record](#)

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop

(14-16 February 2006, UNLV, Las Vegas) source record

[5] — Tachiev & Froese Fischer 2002, A&A 385, 716 DOI: 10.1051/0004-6361/20011816. source record

Model and grid sources

[6] — Amarsi et al. 2020, A&A 642 A62 DOI: 10.1051/0004-6361/202038650. source record

[7] — Amarsi, Grevesse, Grumer, Asplund, Barklem & Collet 2020, A&A 636, A120 DOI: 10.1051/0004-6361/202037890. source record

[8] — Amarsi 'Grid/NLTE' Zenodo, version 3 (the original 13-element 2020 release) DOI: 10.5281/zenodo.3982506. source record

[9] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

Molecular line data and delivery

[10] — Brooke et al. 2014, ApJS 210, 23 DOI: 10.1088/0067-0049/210/2/23. source record

[11] — Sneden, C.; Lucatello, S.; Ram, R. S.; Brooke, J. S. A.; Bernath, P. (2014). Line Lists for the Red and Violet Systems of CN, $^{13}\text{C}^{14}\text{N}$, and $^{12}\text{C}^{15}\text{N}$, and Application to Astronomical Spectra. ApJS 214, 26. DOI: 10.1088/0067-0049/214/2/26. source record